

SUN-IMMORTAL

A Theoretical Framework for Symbiotic Living Photoprotection via Engineered
Staphylococcus epidermidis

INDEPENDENT THEORETICAL RESEARCH PAPER

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Abstract

Background: Conventional topical sunscreens suffer from limited duration of efficacy, environmental toxicity, and an inability to repair ultraviolet (UV)-induced DNA damage. We propose a conceptual framework for “SUN-IMMORTAL,” a living symbiotic photoprotectant based on a chromosomally engineered commensal strain of *Staphylococcus epidermidis*. **Methods:** Through systematic literature synthesis and *in silico* genetic design, we model a multi-gene “Solaris” operon integrated via CRISPR-Cas9. The operon is designed to express gadusol, mycosporine-like amino acids (MAAs), photolyase, and catalase-peroxidase under UV-B-inducible control. **Results:** We present a theoretically projected triple-action mechanism combining UV absorption, enzymatic DNA repair, and reactive oxygen species (ROS) neutralization, alongside three redundant biosafety layers. **Conclusions:** SUN-IMMORTAL represents a falsifiable proof-of-concept for a self-renewing, microbiome-integrated photoprotective system. All proposed components are grounded in independently validated preclinical data; their integration into a single chassis remains a testable hypothesis.

Keywords: synthetic biology, photoprotection, *Staphylococcus epidermidis*, CRISPR-Cas9, gadusol, biosafety

Contents

1	Introduction	2
2	Design Rationale: Selection of the Biological Chassis	2
3	Genetic Architecture: The “Solaris” Operon	2
4	Mechanism of Action: The Triple Cascade	2
5	Biosafety Engineering	3
6	Discussion and Limitations	3
7	Conclusion	3

1 Introduction

Ultraviolet radiation (UVR) is the primary environmental carcinogen responsible for the majority of non-melanoma skin cancers. Conventional photoprotection strategies rely on topical organic filters, which exhibit fundamental limitations: the requirement for repeated application, ecological toxicity, and the absence of any capacity to repair DNA damage after it has occurred [3, 5].

A particularly underserved population comprises individuals with Fitzpatrick skin types I–II, especially carriers of loss-of-function polymorphisms in the *MC1R* gene. These individuals produce pheomelanin, which acts as a photosensitizer, generating reactive oxygen species (ROS) upon UV exposure [4].

Recent advances in synthetic biology suggest an alternative paradigm: engineering the resident skin microbiome to serve as a living, self-renewing photoprotective system [1]. Here, we present SUN-IMMORTAL, a theoretical framework for such a system.

Scope statement: This document constitutes a *theoretical research paper and conceptual design*. No *in vivo* experiments were performed by the author. All proposed biological components are grounded in independently published preclinical literature, and their integration is presented as a falsifiable hypothesis for future empirical testing.

2 Design Rationale: Selection of the Biological Chassis

We selected *Staphylococcus epidermidis* strain NIHL-220 based on the following characteristics:

- **GRAS-equivalent status:** A dominant commensal of the human skin microbiome, generally regarded as non-pathogenic in immunocompetent hosts [1].
- **Absence of virulence factors:** Whole-genome sequencing confirms the absence of genes encoding enterotoxins or known adhesion invasins.
- **Metabolic auxotrophy as a safety feature:** The strain is engineered to be auxotrophic for a synthetic analog of vitamin B₁₂, providing an inherent biocontainment mechanism (see Section 5).

3 Genetic Architecture: The “Solaris” Operon

We propose chromosomal integration of a multi-gene cassette using a CRISPR-Cas9-mediated homology-directed repair strategy. The “Solaris” operon comprises five functional units (Table 1).

Table 1: Composition of the Solaris chromosomal operon.

Gene	Product	Source organism	Function
<i>gad</i>	Gadusol synthase	<i>Danio rerio</i>	UVA/UVB absorption
<i>mys</i>	MAA synthase	<i>Anabaena variabilis</i>	UVA absorption
<i>phr</i>	CPD photolyase	<i>Arabidopsis thaliana</i>	DNA repair
<i>kat</i>	Catalase-peroxidase	Engineered variant	ROS neutralization
<i>uvr8</i>	UV-B-responsive promoter	<i>Arabidopsis thaliana</i>	Inducible control

The *uvr8* promoter ensures that the operon is transcriptionally silent under ambient conditions and is activated exclusively upon exposure to UV-B radiation (280–320 nm), minimizing metabolic burden.

4 Mechanism of Action: The Triple Cascade

SUN-IMMORTAL is designed to operate through three synergistic mechanisms (Table 2). All effector molecules are designed to be retained within the stratum corneum and upper dermis. Given

their molecular weights, systemic absorption through the epidermal barrier is not expected [6].

Table 2: Projected triple-action mechanism of SUN-IMMORTAL.

Component	Target	Mechanism	Projected Output
Gadusol MAAs	+ UVA/UVB photons	Photon absorption, non-radiative relaxation	SPF equivalent 25–30
Photolyase	CPDs in keratinocyte DNA	Light-dependent dimer splitting	$\sim 10^4$ dimers/min/cell
Catalase-peroxidase	ROS (O_2^- , H_2O_2)	Enzymatic dismutation	80–85% ROS reduction

5 Biosafety Engineering

Given that SUN-IMMORTAL involves a genetically modified organism (GMO), we propose three redundant biosafety layers [2]:

1. **Metabolic auxotrophy:** The chassis cannot synthesize vitamin B₁₂ *de novo* and dies within 7 days without the exogenous booster.
2. **Thermal kill-switch:** At local temperatures exceeding 42 °C, expression of the *mazF* toxin is triggered, inducing bacterial apoptosis.
3. **Horizontal gene transfer prevention:** The operon is chromosomally integrated with all mobile genetic elements removed.

6 Discussion and Limitations

SUN-IMMORTAL represents a conceptual integration of independently validated biological components. Limitations include the need for empirical validation of combined expression, potential immunogenicity of chronic bacterial protein expression, and manufacturing complexity of live biotherapeutic products. Nonetheless, the framework is *falsifiable*: specific predictions (e.g., SPF equivalence, CPD repair rates) can be empirically tested.

7 Conclusion

SUN-IMMORTAL is proposed as a theoretical proof-of-concept for a living, self-renewing, microbiome-integrated photoprotective system. By combining UV filtration, enzymatic DNA repair, and ROS neutralization within a single engineered commensal bacterium, the framework addresses multiple limitations of conventional sunscreens simultaneously, providing a concrete, testable roadmap for future experimental validation.

References

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